Global Well-Posedness, Neutral-State Rejection, and Ensemble-Induced Shell Coexistence in a Regularized Multicomponent Brazovskii Functional

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Draft v0.1.40 – 2026-09-04

Abstract

We study one explicitly defined three-component complex field functional on a fixed periodic three-torus. The field is treated as six real components and the derivative Class-II terms are regularized by a positive density floor. For this explicitly defined functional, we prove global well-posedness of its real-L^2 gradient flow for every H^2 datum, exact energy dissipation, continuous dependence, and positive-time smoothness. An exact completion of the quadratic and local polynomial terms, together with a positive radial derivative of the Class-II energy, shows that the unconstrained functional has the zero field as its unique critical point and global minimizer and that the gradient flow decays exponentially in L^2. We then solve a distinct, imposed ensemble problem. On the side-16 torus, the exact spectral bottom lies on the integer shell $|n|^2 = 3$. After fixing $Q = \|\Psi\|_2^2/2$ or introducing $\mathcal{O}_\mu = \mathcal{F} - \mu Q$, an exact nonnegative decomposition gives a finite-shell constrained minimizer and a first-order grand-potential coexistence point. The charge or chemical potential is imposed rather than derived; the paper therefore makes no claim about a physical vacuum, BCC structure, infinite volume, or a quantum continuum limit. We provide a claim-to-proof map and a reproducibility protocol tied to the repository's independent exact audits.

1 Introduction and statement of scope

There are two mathematically different questions in the current TECT/A2 candidate. The first is whether a fixed regularized functional defines a well-behaved dissipative evolution and what its unconstrained equilibria are. The second is what happens after the variational problem is changed by fixing a nonzero quadratic charge or by subtracting a chemical-potential term. A nonzero minimizer of the second problem must not be reported as an equilibrium of the first problem. This distinction is the organizing principle of the present paper.

The object is a specific functional, not an assertion that this functional is the physical law of nature. Its identification with the canonical P1 production functional is retained as the named hypothesis `A2-H3-CANONICAL-PRODUCTION-FUNCTIONAL`, but that hypothesis is not used as an analytic premise below. All theorems are statements about the explicitly defined mathematical functional; if H3 is independently accepted, the results also transfer to the canonical P1 interpretation. Because the paper fixes the raw Laplacian convention, the unresolved shorthand in the later A2 summary blocks only canonical transfer, not the standalone theorem.

The paper has three main contributions.

(i) We give a continuum proof for the full regularized three-component fourth-order functional, including the derivative Class-II terms. The nonlinear map has order two, not four, and the endpoint H^4 estimate is obtained by a Duhamel cancellation rather than by an invalid fractional-smoothing shortcut.

(ii) We give an exact global sign test for the pinned unconstrained problem. It rejects every nonzero equilibrium of this candidate, not merely the finite list of lattice patterns tested previously.

(iii) We solve the separately declared fixed-charge and grand-canonical problems by an exact spectral and polynomial completion. The result is a finite-volume coexistence mechanism with an explicit saturated-state charge Q_* at the coexistence value, while the original neutral reference remains lower. We do not identify the charge of every global minimizer away from coexistence.

The statements are deliberately narrower than a constructive quantum-field theory theorem. We do not claim an infinite-volume limit, a real-time automorphism group, algebraic KMS dynamics, a ground-state gap, a continuum renormalization, a derived conserved charge, a density crystal, BCC selection, or any cosmological interpretation.

2 The functional and conventions

Let $\mathbb{T}^3_{16} = [0,16)^3$ with periodic boundary conditions and let $\Psi : \mathbb{T}^3_{16} \to \mathbb{C}^3$. We regard $\mathbb{C}^3$ as $\mathbb{R}^6$ and use the real pairing

$$\langle U, V\rangle_{\mathbb{R}} = \operatorname{Re} \int_{\mathbb{T}^3_{16}} U^\dagger V \, \mathrm{d}x. \tag{2.1}$$

For a periodic Fourier series, set

$$\|U\|^2_{H^2_\Delta} := \sum_{n\in\mathbb{Z}^3} (1 + |k_n|^4)|\widehat{U}(n)|^2_{\mathbb{C}^3} = \|U\|_2^2 + \|\Delta U\|_2^2, \qquad k_n = (2\pi/16)n. \tag{2.2}$$

This graph norm is equivalent to the standard H^2 norm on the fixed torus; all H^2 assertions below are unchanged by this equivalence. Write $\rho = \Psi^\dagger\Psi$. For the three Hermitian internal generators T_A displayed below, set

$$m_A = \Psi^\dagger T_A \Psi, \qquad J_A = \nabla m_A, \qquad K_A = J_A - \frac{m_A}{\rho + \varepsilon_\rho}\nabla\rho, \qquad \varepsilon_\rho = 10^{-12}. \tag{2.3}$$

For complete specification, the three Hermitian generators (the Pauli basis embedded in the first two internal components) are

$$T_1 = \begin{pmatrix} 0 & 1 & 0\\ 1 & 0 & 0\\ 0 & 0 & 0\end{pmatrix}, \quad T_2 = \begin{pmatrix} 0 & -i & 0\\ i & 0 & 0\\ 0 & 0 & 0\end{pmatrix}, \quad T_3 = \begin{pmatrix} 1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & 0\end{pmatrix}. \tag{2.4}$$

The positive floor in (2.3) is part of the declared functional; removing it is a different theorem.

The quadratic operator is

$$L = (r + Z(-\Delta) + Y(-\Delta)^2)\mathrm{I}_3 + M_{\mathrm{fam}} + k_{\mathrm{lock}}(\mathrm{I}_3 - P_0), \tag{2.5}$$

where the pinned scalar coefficients are

$$Y = 1, \qquad Z = -0.9252754126, \qquad r = 0.4740336473, \tag{2.6}$$

The remaining pinned internal data are

$$M_{\mathrm{fam}} = \operatorname{diag}\left(0, \frac{3}{100}, \frac{7}{100}\right), \quad z_0 = (1,1,1)^T, \quad k_{\mathrm{lock}} = \frac{3}{20}, \quad P_0 = \frac{z_0 z_0^\dagger}{z_0^\dagger z_0} = \frac{1}{3}\begin{pmatrix} 1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{pmatrix}. \tag{2.7}$$

The shell-bias coefficient is fixed to $\eta_{\text{shell}} = 0$, so no additional Fourier-shell term is present. The sum of the family and lock contributions is the explicit internal matrix displayed in (5.2). The family/lock contribution is therefore positive semidefinite. The local polynomial coefficients are

$$\lambda = -\frac{43}{100}, \qquad \gamma = \frac{81}{50} > 0. \tag{2.8}$$

The Class-II coefficients are fixed by

$$\begin{aligned}
&\varepsilon_M = 10^{-12}, \quad \alpha_X = \frac{3}{10}, \quad \beta_X = \frac{1}{4}, \quad M_X = 2,\\
&c_{JJ} = \frac{1}{5}, \quad c_{JK} = \frac{1}{10}, \quad c_{KK} = \frac{3}{20},\\
&a = \frac{c_{JJ}\alpha_X^2}{M_X^2 + \varepsilon_M}, \qquad b = \frac{c_{JK}\alpha_X\beta_X}{M_X^2 + \varepsilon_M},\\
&c = \frac{c_{KK}\beta_X^2}{M_X^2 + \varepsilon_M}.
\end{aligned} \tag{2.9}$$

For orientation, these formulas give approximately $a = 4.5 \times 10^{-3}$, $b = 1.875 \times 10^{-3}$, and $c = 2.34375 \times 10^{-3}$; the displayed formulas, rather than rounded numerical values, define the functional. The Class-II density is

$$\mathcal{F}_{\text{II}}(\Psi) = \sum_A \int_{\mathbb{T}^3_{16}} \left(\frac{a}{2}|J_A|^2 + b\,\mathrm{Re}(J_A^\dagger K_A) + \frac{c}{2}|K_A|^2 \right) \mathrm{d}x, \tag{2.10}$$

with the pinned positive-definite coefficient matrix $Q_{\text{II}} = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$. The full functional is

$$\mathcal{F}(\Psi) = \frac{1}{2}\langle \Psi, L\Psi\rangle_{\mathbb{R}} + \int_{\mathbb{T}^3_{16}} \left(\frac{\lambda}{4}\rho^2 + \frac{\gamma}{6}\rho^3 \right) \mathrm{d}x + \mathcal{F}_{\text{II}}(\Psi). \tag{2.11}$$

The domain for the evolution theorem is $H^2(\mathbb{T}^3_{16}; \mathbb{C}^3)$, while the linear operator has domain H^4 and form domain H^2. All assertions about the charge ensemble below refer to the same fixed torus and the same functional (2.11); no new physical field or reservoir is introduced.

3 Main results

Theorem 3.1 (full gradient-flow well-posedness)**.** *For every $\Psi_0 \in H^2(\mathbb{T}^3_{16}; \mathbb{C}^3)$, the real-$L^2$ gradient flow*

$$\partial_t \Psi = -\nabla_{\mathbb{R}}\mathcal{F}(\Psi), \qquad \Psi(0) = \Psi_0, \tag{3.1}$$

has a unique global solution with

$$\Psi \in C([0,\infty); H^2) \cap L^2_{\text{loc}}((0,\infty); H^4), \qquad \partial_t\Psi \in L^2(0,T;L^2) \quad (0 < T < \infty). \tag{3.2}$$

The solution depends continuously on the initial datum in H^2 on every finite time interval and satisfies

$$\mathcal{F}(\Psi(t)) + \int_s^t \|\partial_\tau \Psi(\tau)\|_2^2 \,\mathrm{d}\tau = \mathcal{F}(\Psi(s)), \qquad 0 \le s \le t. \tag{3.3}$$

For every $t > 0$ it is smooth in space and time.

Theorem 3.2 (neutral minimizer and exponential decay)**.** *For the pinned coefficients, there are exact rational constants*

$$g = \frac{7198187500255823388837}{5400000000000000000000} > \frac{1}{8}, \qquad \kappa = \frac{21016750000767470165111}{81000000000000000000000} > \frac{1}{4} \tag{3.4}$$

such that every $\Psi \in H^2$ obeys

$$\mathcal{F}(\Psi) \geq g \, \|\Psi\|_2^2 \,, \qquad \langle D\mathcal{F}(\Psi), \Psi\rangle_{\mathbb{R}} \geq \kappa \, \|\Psi\|_2^2 \,. \tag{3.5}$$

Consequently $\Psi = 0$ is the unique critical point and global minimizer of the unconstrained problem, and every solution of (3.1) obeys

$$\|\Psi(t)\|_2^2 \leq e^{-2\kappa t} \, \|\Psi_0\|_2^2 \,. \tag{3.6}$$

Theorem 3.3 (imposed charge and chemical-potential ensemble)**.** *Let $Q(\Psi) = \|\Psi\|_2^2 /2$ and let λ_0 be the exact bottom of the quadratic operator (2.5) on the side-16 torus. Put*

$$\rho_* = \frac{43}{216}, \qquad c_* = \frac{1849}{86400}, \qquad \mu_t = \lambda_0 - c_*. \tag{3.7}$$

Then, with $\rho = \Psi^\dagger \Psi$,

$$\mathcal{F}(\Psi) = \mu_t Q(\Psi) + \frac{1}{2} \langle \Psi, (L - \lambda_0)\Psi\rangle_{\mathbb{R}} + \mathcal{F}_{\mathrm{II}}(\Psi) + \frac{\gamma}{6} \int_{\mathbb{T}^3_{16}} \rho(\rho - \rho_*)^2 \, \mathrm{d}x. \tag{3.8}$$

Every term after $\mu_t Q$ is nonnegative. A ground-shell plane wave with constant density ρ_* is a constrained global minimizer at

$$Q_* = \frac{11008}{27}. \tag{3.9}$$

For $\mathcal{O}_\mu = \mathcal{F} - \mu Q$, zero is the unique minimizer for $\mu < \mu_t$, coexists with nonzero ground-shell minimizers at $\mu = \mu_t$, and is beaten by a nonzero minimizer for $\mu > \mu_t$. The coexistence point lies strictly between the amplitude saddle-node and the zero-field spinodal:

$$\mu_{\mathrm{sn}} = \lambda_0 - \frac{1849}{64800} < \mu_t = \lambda_0 - \frac{1849}{86400} < \lambda_0 = \mu_{\mathrm{lin}}. \tag{3.10}$$

This is a theorem about an imposed ensemble, not a derived physical charge or neutral-vacuum selection.

4 Proof of the evolution theorem

4.1 Linear operator and coercivity

The scalar symbol of (2.5) is $p(x) = r + Zx + Yx^2$ with $x = |k|^2$. Here $Y > 0$ and $Z < 0$, so the vertex lies in $[0, \infty)$. Completing the square therefore gives the exact continuous-shell bound

$$p(x) = Y \left(x + \frac{Z}{2Y} \right)^2 + r - \frac{Z^2}{4Y}, \qquad \inf_{x \geq 0} p(x) = \frac{26000000000947494031}{10^{20}} > 0. \tag{4.1}$$

For an explicit Sobolev-equivalent constant, subtract one fifth of the graph Fourier weight $1+x^2$. With the pinned finite-decimal coefficients,

$$p(x) - \frac{1}{5}(1+x^2) = \frac{4}{5}x^2 - \frac{4626377063}{5000000000}x + \frac{2740336473}{10000000000}, \qquad \mathrm{disc} = -\frac{519327054947494031}{25000000000000000000} < 0. \tag{4.2}$$

The leading coefficient is positive, hence this polynomial is positive for all real x and

$$\frac{p(x)}{1 + x^2} \geq \frac{1}{5} \quad (x \geq 0). \tag{4.3}$$

The family matrix and the lock term are positive semidefinite. For completeness, on Fourier coefficients the operator is the multiplier $\widehat{LU}(n) = \ell_n \widehat{U}(n)$, with $\ell_n = p(|k_n|^2)\mathrm{I}_3 + M_{\mathrm{fam}} + k_{\mathrm{lock}}(\mathrm{I}_3 - P_0)$. Each ℓ_n is Hermitian and, for constants $c, C > 0$ independent of n,

$$c(1 + |k_n|^4)\mathrm{I}_3 \leq \ell_n \leq C(1 + |k_n|^4)\mathrm{I}_3.$$

The diagonal Fourier multiplier with domain $\{U : \sum_n (1+|k_n|^8)|\widehat{U}(n)|^2 < \infty\} = H^4$ is therefore self-adjoint and positive; its quadratic-form domain is $D(L^{1/2}) = H^2$, and the displayed graph and Sobolev norms are equivalent. The pinned Class-II matrix is positive definite because its coefficient definitions give the exact factorisation

$$ac - b^2 = \frac{\alpha_X^2 \beta_X^2}{(M_X^2 + \varepsilon_M)^2}(c_{JJ}c_{KK} - c_{JK}^2) = \frac{\alpha_X^2 \beta_X^2}{(M_X^2 + \varepsilon_M)^2} \frac{1}{50} > 0. \tag{4.4}$$

Since $\lambda < 0$ and $\gamma > 0$, Young's inequality gives

$$\frac{|\lambda|}{4}\rho^2 \leq \frac{\gamma}{12}\rho^3 + C_Y, \qquad C_Y = \frac{|\lambda|^3}{3\gamma^2} = \frac{79507}{7873200}. \tag{4.5}$$

Combining (4.3), (4.4), and (4.5) yields with the explicit choice $c_{H^2_\Delta} = 1/5$,

$$\mathcal{F}(\Psi) \geq \frac{c_{H^2_\Delta}}{2} \|\Psi\|^2_{H^2_\Delta} + \frac{\gamma}{12} \|\Psi\|^6_{L^6} - C_Y|\mathbb{T}^3_{16}|. \tag{4.6}$$

4.2 The Class-II map has order two

Realify Ψ as $u \in \mathbb{R}^6$. Let A_A be the real symmetric realification of T_A, and define

$$p_A(u) = 2A_A u, \qquad q_A(u) = \frac{u^T A_A u}{u^T u + \varepsilon_\rho}, \qquad s_A(u) = 2(A_A - q_A(u)\mathrm{I})u. \tag{4.7}$$

The Class-II density can then be written exactly as

$$G_{\mathrm{II}}(u, \nabla u) = \frac{1}{2}\sum_j (\partial_j u)^T B(u)(\partial_j u), \quad B(u) = \sum_A \{a p_A p_A^T + b(p_A s_A^T + s_A p_A^T) + c s_A s_A^T\}. \tag{4.8}$$

Here $\nabla^2 u$ denotes the raw componentwise Laplacian $\Delta u := \sum_{j=1}^3 \partial_j^2 u$; the positive elliptic operator in L is $-\Delta$. The positive ε_ρ makes B smooth on all of $\mathbb{R}^6$. Direct Euler–Lagrange differentiation therefore has the form

$$N_{\mathrm{II}}(u) = -B(u)\nabla^2 u + C(u)[\nabla u, \nabla u], \tag{4.9}$$

where C is smooth and no derivative above order two occurs. More explicitly, with $\partial_\alpha = \partial/\partial u_\alpha$ and summation over repeated Greek and spatial indices,

$$[N_{\mathrm{II}}(u)]_\alpha = \frac{1}{2}\,\partial_\alpha B_{\beta\gamma}(u)\,\partial_j u_\beta\,\partial_j u_\gamma - \partial_j(B_{\alpha\gamma}(u)\,\partial_j u_\gamma)\,, \qquad C_{\alpha\delta\gamma}(u) = \frac{1}{2}\partial_\alpha B_{\delta\gamma}(u) - \partial_\delta B_{\alpha\gamma}(u). \tag{4.10}$$

The minus sign in the leading term is therefore the integration-by-parts sign from the Dirichlet part of the Class-II density. The earlier A2 v1.1 note prints the same negative component formula, whereas the later integrated A2 v2.0 note contains a schematic positive $B(u)\nabla^2 u$ in its corresponding summary equation. A constant-coefficient check with $B = I$ gives $\delta(\frac{1}{2}\int |\nabla u|^2) = -\Delta u$. Thus the two displays are notation-compatible only if the v2.0 symbol ∇^2 means the positive operator $-\Delta$; if it means the raw componentwise Δ, its sign is opposite. Because v2.0 does not define ∇^2 in that equation, 'EXP-001388' narrows but does not close the source-reconciliation gate. An independent paper-local source/sign audit is provided in `classii_sign_audit.py` under the verification directory. It reads both hash-pinned notes and repeats the one-mode sign test with exact rational arithmetic. Its artifact `classii-sign.json` records an 8/8 pass and both source hashes. This is an audit certificate, not a canonical-source correction or an external reconciliation. The paper convention is the raw Δ defined above, fixed by the displayed density and real pairing. The source discrepancy is retained as an open canonical-transfer gate rather than silently changed; it is not an additional hypothesis in the standalone theorem. To fix the notation used in the evolution equation, let the real gradient of the local polynomial be

$$N_{\mathrm{loc}}(u) = \lambda(u^T u)u + \gamma(u^T u)^2 u, \qquad N(u) = N_{\mathrm{loc}}(u) + N_{\mathrm{II}}(u). \tag{4.11}$$

The local term is order zero and is locally Lipschitz from H^2 to L^2; therefore the estimate below is for the full lower-order map N, not only its Class-II part. On the fixed three-torus, $H^2 \hookrightarrow L^\infty \cap W^{1,6}$ and $L^3 \hookrightarrow L^2$. For $\|u\|_{H^2}, \|v\|_{H^2} \le R$, smoothness of the regularized coefficient maps gives

$$\|B(u) - B(v)\|_\infty + \|C(u) - C(v)\|_\infty \le C_R \|u - v\|_{H^2}. \tag{4.12}$$

Using $\Delta u \in L^2$, $\nabla u \in L^6$, and the finite-volume embedding $L^3 \hookrightarrow L^2$, the displayed order-two formula then yields

$$\|N_{\mathrm{II}}(u) - N_{\mathrm{II}}(v)\|_2 \le C_R\big[(1 + \|\Delta u\|_2) \|u - v\|_{H^2} + (\|\nabla u\|_6 + \|\nabla v\|_6) \|\nabla u - \nabla v\|_6\big] \tag{4.13}$$

$$\le C_R \|u - v\|_{H^2}. \tag{4.14}$$

The algebra property of H^2 gives the same estimate for N_{loc}, and hence

$$\|N(u) - N(v)\|_2 \le C_R \|u - v\|_{H^2} \tag{4.15}$$

on every bounded H^2 ball. This is the product estimate used by the contraction argument; no higher spatial derivative is hidden in N.

The positive fourth-order operator generates an analytic semigroup [1, 2]. This can also be seen directly from the multiplier. If $\{\ell_{n,j}\}$ are the eigenvalues of the Hermitian mode matrices, then $c(1 + |k_n|^4) \le \ell_{n,j} \le C(1 + |k_n|^4)$, and for $0 < \alpha \le 1$,

$$\left\|L^\alpha e^{-tL}\right\|_{L^2 \to L^2} = \sup_{n,j} \ell_{n,j}^\alpha e^{-t\ell_{n,j}} \le \left(\frac{\alpha}{et}\right)^\alpha, \qquad D(L^\alpha) = H^{4\alpha}. \tag{4.16}$$

The case $\alpha = 1/2$ is the estimate

$$\left\|L^{1/2} e^{-tL}\right\|_{L^2 \to L^2} \le C t^{-1/2}, \qquad t > 0, \tag{4.17}$$

whose kernel is integrable at the origin. To expose the local-existence argument, set $X_T = C([0, T]; H^2)$ and

$$(\mathcal{T} u)(t) = e^{-tL} u_0 - \int_0^t e^{-(t-s)L} N(u(s))\, \mathrm{d}s. \tag{4.18}$$

On an X_T ball of H^2 radius R, (4.15), $N(0) = 0$, and (4.17) give

$$\begin{aligned} \sup_{t \le T} \left\|\mathcal{T} u(t) - e^{-tL} u_0\right\|_{H^2} &\le C_R T^{1/2}, \\ \|\mathcal{T} u - \mathcal{T} v\|_{X_T} &\le C_R T^{1/2} \|u - v\|_{X_T}. \end{aligned} \tag{4.19}$$

Strong continuity of e^{-tL} on H^2 lets one first fix the ball around u_0 and then choose T so that (4.19) maps it into itself with contraction constant below one. Banach's theorem gives the unique local mild solution. Reapplying the same construction on an H^2-bounded terminal ball gives the continuation alternative: a finite maximal time can occur only if the H^2 norm diverges. Thus the semigroup references [1, 2] provide background, while the estimates used here are the displayed multiplier and fixed-point bounds.

4.3 Galerkin passage and energy identity

Let P_n be the real L^2 Fourier projection, set $u_n(0) = P_n u_0$, and solve

$$\partial_t u_n = -P_n(Lu_n + N(u_n)). \tag{4.20}$$

This is the exact finite-dimensional real gradient flow. Moreover, $P_n u_0 \to u_0$ in H^2 and the displayed energy is continuous on bounded H^2 sets, so $\mathcal{F}(P_n u_0) \to \mathcal{F}(u_0)$. Hence

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}(u_n) = -\left\|\partial_t u_n\right\|_2^2. \tag{4.21}$$

The convergent initial energies and coercivity estimate give a uniform $L^\infty(0,T;H^2)$ bound, while (4.21) gives an $L^2(0,T;L^2)$ bound on $\partial_t u_n$. The common H^2 ball, $N(0) = 0$, and (4.15) also give a uniform $L^\infty(0,T;L^2)$ bound on $N(u_n)$. Ellipticity applied to $Lu_n = -\partial_t u_n - P_n N(u_n)$ then gives an $L^2(0,T;H^4)$ bound. For completeness, the required compactness can be seen directly rather than left to the name of the Aubin–Lions–Simon lemma [17]. Let Q_K be the real Fourier projection onto modes with $|k_n| \le K$. The uniform $L^2(0,T;H^4)$ bound gives the high-frequency estimate

$$\left\|(\mathrm{I} - Q_K)u_n\right\|^2_{L^2(0,T;H^2)} \le C(1+K^2)^{-2}\left\|u_n\right\|^2_{L^2(0,T;H^4)}. \tag{4.22}$$

For fixed K, the derivative bound makes $Q_K u_n$ bounded in $H^1(0,T;Q_K L^2)$. Because $Q_K L^2$ is finite dimensional, the scalar one-dimensional Rellich theorem applied to its finitely many coefficients makes this sequence precompact in $L^2(0,T;Q_K L^2)$; all Sobolev norms are equivalent on that range. First choose K so that (4.22) is uniformly small and then take a convergent subsequence of the low modes. The resulting diagonal argument gives strong convergence in $L^2(0,T;H^2)$. Equation (4.15) and the common $L^\infty(0,T;H^2)$ ball then give strong convergence of the nonlinear term in $L^2(0,T;L^2)$. The weak limit satisfies the Galerkin equation in H^{-2}, and the weak $L^2(0,T;H^4)$ bound passes to the limit. Since the local Lipschitz estimate gives $N(u) \in L^2(0,T;L^2)$, the limit equation upgrades to

$$u \in L^2(0,T;H^4) \cap H^1(0,T;L^2), \qquad Lu + N(u) \in L^2(0,T;L^2). \tag{4.23}$$

This upgrade is the time-regularity input used below; it is not inferred from fractional smoothing alone. Since the estimate holds on the full interval $(0,T)$, it also supplies the endpoint integrability needed to read (3.3) at $s = 0$.

The continuity assertion at the Hilbert-scale midpoint is also explicit. For a positive self-adjoint L, spectral truncation gives, whenever $w \in L^2(0,T;D(L)) \cap H^1(0,T;L^2)$,

$$\begin{aligned}&w \in C([0,T];D(L^{1/2})),\\ &\frac{1}{2}\left\|L^{1/2}w(t)\right\|_2^2 - \frac{1}{2}\left\|L^{1/2}w(s)\right\|_2^2 = \int_s^t \langle Lw(\tau), \partial_\tau w(\tau)\rangle_{\mathbb{R}}\,\mathrm{d}\tau.\end{aligned} \tag{4.24}$$

Indeed, the identity is the ordinary finite-dimensional chain rule for $P_M w$; $LP_M w \to Lw$ and $P_M \partial_t w \to \partial_t w$ in $L^2(0,T;L^2)$, and the resulting absolutely continuous midpoint norm, together with weak midpoint continuity, gives strong continuity in $D(L^{1/2})$. Applying this to $w = u$ and using $D(L^{1/2}) = H^2$ supplies the claimed $C([0,T];H^2)$ representative, preserves $u(0) = u_0$, and proves the quadratic chain rule including $s = 0$.

For clarity, the nonlinear energy is

$$\Phi(u) = \int_{\mathbb{T}^3_{16}} \left(\frac{\lambda}{4}(u^T u)^2 + \frac{\gamma}{6}(u^T u)^3\right)\mathrm{d}x + \mathcal{F}_{\mathrm{II}}(u). \tag{4.25}$$

Its derivative identity $D\Phi(u)[h] = \langle N(u), h\rangle_{\mathbb{R}}$ and the local Lipschitz estimate are understood in the Gelfand triple $H^2 \hookrightarrow L^2 \hookrightarrow H^{-2}$. In particular, $D\Phi : H^2 \to L^2 \subset H^{-2}$ is locally Lipschitz.

The previous upgrade gives, on every finite interval, both $u \in L^2(s,t;H^4)$ and $\partial_t u \in L^2(s,t;L^2)$. Consequently the L^2 pairing below is well-defined; the Gelfand triple is used to identify the weak limit, not to pair two H^{-2} quantities. For completeness, the chain rule does not require pairing two distributions in H^{-2}. Project u onto the first M real Fourier modes and apply a time mollifier of width ε; the resulting finite-dimensional smooth curve satisfies the ordinary chain rule. First let $\varepsilon \downarrow 0$ and then $M \to \infty$. The convergences $P_M u \to u$ in $L^2(s,t;H^2)$ and $P_M \partial_t u \to \partial_t u$ in $L^2(s,t;L^2)$, together with the bounded-set Lipschitz estimate for N, give

$$\int_s^t \langle N(P_M u), \partial_\tau P_M u\rangle_{\mathbb{R}}\, \mathrm{d}\tau \longrightarrow \int_s^t \langle N(u), \partial_\tau u\rangle_{\mathbb{R}}\, \mathrm{d}\tau. \tag{4.26}$$

The uniform H^2 bound controls the mollified curves, so this limit also justifies the C^1 energy chain rule at the endpoints by continuity in H^2. The Bochner/Hilbert chain rule for the C^1 energy Φ on bounded H^2 sets therefore gives

$$\Phi(u(t)) - \Phi(u(s)) = \int_s^t \langle N(u(\tau)), \partial_\tau u(\tau)\rangle_{\mathbb{R}}\, \mathrm{d}\tau. \tag{4.27}$$

The quadratic part is $\frac{1}{2}\left\|L^{1/2}u\right\|_2^2$ and obeys the explicit Hilbert-scale identity (4.24). Combining it with (4.27) proves (3.3), not just an energy inequality. The lower bound (4.6) prevents the H^2 norm from diverging at a finite maximal time, so the solution is global.

4.4 Continuous dependence and smoothing

For two solutions in a common energy-bounded ball, the mild difference equation has the Volterra estimate

$$\|u(t) - v(t)\|_{H^2} \le C\,\|u_0 - v_0\|_{H^2} + C_R \int_0^t (t-s)^{-1/2}\,\|u(s) - v(s)\|_{H^2}\, \mathrm{d}s. \tag{4.28}$$

This weakly singular step also reduces to ordinary Gronwall. With $k(t) = t^{-1/2}$ and $d(t) = \|u(t) - v(t)\|_{H^2}$, iterate (4.28) once. Since $(k*k)(t) = \int_0^t (t-s)^{-1/2} s^{-1/2}\, \mathrm{d}s = \pi$, one obtains on $[0,T]$

$$d(t) \le C_T\,\|u_0 - v_0\|_{H^2} + \pi C_R^2 \int_0^t d(s)\, \mathrm{d}s. \tag{4.29}$$

Ordinary Gronwall therefore proves uniqueness and continuous dependence on every finite interval. Analytic-semigroup estimates first give $u(t) \in D(L^\alpha) = H^{4\alpha}$ for $t > 0$ and every $\alpha < 1$. We make the time modulus used below explicit. Fix $0 < a < b$ and choose $\alpha \in (1/2, 1)$. The mild formula, boundedness of $f = N(u)$ in L^2 on $[a,b]$, and the fractional multiplier estimate give

$$\sup_{t\in[a,b]} \|u(t)\|_{D(L^\alpha)} \le C_{a,b,\alpha}. \tag{4.30}$$

The time-regularity upgrade above also gives, for $s,t \in [a,b]$,

$$\|u(t) - u(s)\|_2 \le \|\partial_t u\|_{L^2(a,b;L^2)}\, |t-s|^{1/2}. \tag{4.31}$$

Set $\vartheta = (2\alpha)^{-1}$. Interpolation between L^2 and $D(L^\alpha)$, together with $D(L^{1/2}) = H^2$, gives

$$\left\|L^{1/2}w\right\|_2 \le C\,\|w\|_2^{1-\vartheta}\,\|L^\alpha w\|_2^{\vartheta}. \tag{4.32}$$

Applying this to $w = u(t) - u(s)$, using (4.30) and (4.31), and using equivalence of the graph and Sobolev norms, yields

$$\|u(t) - u(s)\|_{H^2} \le C_{a,b,\alpha}|t-s|^{(2\alpha-1)/(4\alpha)}. \tag{4.33}$$

The exponent is positive because $\alpha > 1/2$. Since $N : H^2 \to L^2$ is locally Lipschitz, $f(t) = N(u(t))$ is therefore Hölder continuous into L^2 on every compact interval away from $t = 0$. The endpoint cancellation

$$L \int_a^t e^{-(t-s)L} f(s)\,\mathrm{d}s = (\mathrm{I} - e^{-(t-a)L}) f(t) + \int_a^t Le^{-(t-s)L}[f(s) - f(t)]\,\mathrm{d}s \tag{4.34}$$

has an integrable last term. The $\alpha = 1$ multiplier bound is

$$\|Le^{-rL}\|_{2\to 2} \le Cr^{-1}, \qquad r > 0. \tag{4.35}$$

If $\|f(t) - f(s)\|_2 \le K|t - s|^\theta$ on $[a, b]$ for some $\theta > 0$, then

$$\left\| \int_a^t Le^{-(t-s)L}[f(s) - f(t)]\,\mathrm{d}s \right\|_2 \le CK \int_0^{t-a} r^{\theta-1}\,\mathrm{d}r = \frac{CK}{\theta}(t - a)^\theta. \tag{4.36}$$

Thus the endpoint integral is an actual L^2 limit, not a formal cancellation. It follows that $u(t) \in D(L) = H^4$ for every $t > 0$. Periodic Moser estimates give

$$N : H^m \longrightarrow H^{m-2} \quad \text{locally Lipschitz}, \qquad m \ge 2, \tag{4.37}$$

More precisely, the periodic Leibniz and composition estimates for the smooth maps B, C give, on every bounded H^m ball,

$$\|N(w)\|_{H^{m-2}} \le C_{m,R}(1 + \|w\|_{H^m}), \qquad \|N(w) - N(z)\|_{H^{m-2}} \le C_{m,R}\|w - z\|_{H^m}. \tag{4.38}$$

We now make the shifted-base-space step explicit. For each integer $m \ge 2$, set $X_m = H^{m-2}$. Here the Sobolev space is on $\mathbb{T}^3_{16}$ with values in $\mathbb{C}^3$. The constant-coefficient Fourier multiplier realization of L commutes with the Sobolev multiplier, so its analytic-semigroup bounds transfer to this base space:

$$D_{X_m}(L) = H^{m+2}, \qquad \|Le^{-rL}\|_{X_m\to X_m} \le C_m r^{-1}, \qquad r > 0. \tag{4.39}$$

Let $f \in C^\theta([a, b]; X_m)$ for some $0 < \theta < 1$, and write $v(t) = \int_a^t e^{-(t-s)L} f(s)\,\mathrm{d}s$. Applying the endpoint identity in X_m gives

$$Lv(t) = (\mathrm{I} - e^{-(t-a)L}) f(t) + \int_a^t Le^{-(t-s)L}[f(s) - f(t)]\,\mathrm{d}s. \tag{4.40}$$

The last integral is convergent because its integrand is bounded by $C_m(t - s)^{\theta-1}$. We record the difference estimate used to propagate this regularity. For $h > 0$ and $q > 0$, the multiplier calculus gives

$$\|(e^{-hL} - \mathrm{I})Le^{-qL}\|_{X_m\to X_m} \le C_m \min\{q^{-1}, hq^{-2}\}. \tag{4.41}$$

Indeed, the first bound uses $\|Le^{-qL}\| \le C_m q^{-1}$ and the second uses $e^{-hL} - \mathrm{I} = -\int_0^h Le^{-\sigma L}\,\mathrm{d}\sigma$ and the $L^2 e^{-rL}$ multiplier bound. If $t > s \ge a + \eta$, put $h = t - s$ and split the two endpoint integrals at s. On the common interval, with $q = s - r$,

$$\begin{aligned} &Le^{-(t-r)L}[f(r) - f(t)] - Le^{-(s-r)L}[f(r) - f(s)] \\ &\quad = (e^{-hL} - \mathrm{I})Le^{-qL}[f(r) - f(s)] + Le^{-(t-r)L}[f(s) - f(t)]. \end{aligned}$$

Writing $K = \|f\|_{C^\theta([a,b];X_m)}$ and noting that $[f]_{C^\theta} \le K$, the first term is bounded after integration by

$$K \int_0^{s-a} \min\{q^{\theta-1}, hq^{\theta-2}\}\,\mathrm{d}q \le \left(\frac{1}{\theta} + \frac{1}{1 - \theta}\right) Kh^\theta, \tag{4.42}$$

where the integral is split at $q = h$; the restriction $\theta < 1$ removes the logarithmic endpoint loss that would occur at $\theta = 1$. The second common term is $O(Kh^\theta)$ because $\int_a^s Le^{-(t-r)L}\,\mathrm{d}r =$

$e^{-(t-s)L} - e^{-(t-a)L}$ is uniformly bounded. The tail interval $[s, t]$ is bounded in the same way by $CK \int_0^h q^{\theta-1}\, dq$. For the explicit endpoint terms, the positive distance from a gives the separate estimate

$$\|(e^{-hL} - \mathrm{I})e^{-(s-a)L}\|_{X_m\to X_m} \le C_{m,\eta}h, \qquad s \ge a + \eta, \tag{4.43}$$

so they are also $O(Kh^\theta)$, using $f(t) - f(s) = O(Kh^\theta)$ and $h \le (b - a)^{1-\theta}h^\theta$. Thus

$$\|Lv(t) - Lv(s)\|_{X_m} + \|v(t) - v(s)\|_{X_m} \le C_{a,b,\eta,m,\theta}\|f\|_{C^\theta([a,b];X_m)}|t - s|^\theta. \tag{4.44}$$

The endpoint identity and the preceding estimate first give $Lv \in C([a + \eta, b]; X_m)$. Since $v' = f - Lv$ in X_m there, v is actually Lipschitz in X_m, which supplies the second term in (4.44) (and hence its weaker θ-Holder form). The linear term $e^{-(t-a)L}u(a)$ is smooth with values in H^{m+2} for $t \ge a + \eta$. The graph norm equivalence for $L : X_m \supset H^{m+2} \to X_m$ therefore gives

$$u \in C^\theta([a + \eta, b]; H^{m+2}) \quad \text{whenever} \quad u \in C([a, b]; H^m), \quad N(u) \in C^\theta([a, b]; X_m). \tag{4.45}$$

This is the shifted-base endpoint bootstrap used below, rather than an unqualified change of the underlying base space.

For $m = 2$, the preceding interpolation estimate supplies a positive Hölder exponent for u in H^2, and the local Lipschitz bound $N : H^2 \to L^2 = X_2$ makes $f = N(u)$ Hölder in X_2. Equations (4.40)–(4.45) then produce $u \in C^\theta(H^4)$ on every compact interval away from $t = 0$. If $u \in C^\theta(H^m)$ has been obtained on such an interval, the Moser estimate (4.38) makes $N(u) \in C^\theta(H^{m-2}) = C^\theta(X_m)$, and the same bootstrap gives $u \in C^\theta(H^{m+2})$ after moving to a slightly later compact interval. This induction is valid on every compact interval strictly inside $(0, \infty)$. We run the ladder on even m; interpolation between adjacent even levels supplies any intermediate integer Sobolev order needed below, while the even ladder alone is sufficient for all spatial embeddings. Taking $m = 2, 4, 6, \ldots$ proves $u(t) \in H^{2+2q}$ for every q and every $t > 0$. To make the time regularity explicit, set

$$\mathcal{F}(u) := Lu + N(u). \tag{4.46}$$

For every even integer $r \ge 0$, the multiplier and periodic Moser estimates imply

$$\begin{aligned} &\mathcal{F} : H^{r+4} \longrightarrow H^r \quad \text{is } C^\infty \text{ on bounded sets,} \\ &\|D^jN(u)[h_1, \ldots, h_j]\|_{H^r} \le C_{r,j,R} \prod_{\ell=1}^{j} \|h_\ell\|_{H^{r+2}}, \qquad j \ge 1, \quad \|u\|_{H^{r+2}} \le R. \end{aligned} \tag{4.47}$$

The fourth-order part is linear, and the displayed bound follows by differentiating the smooth regularized coefficient maps and applying the periodic product estimates. Thus the equation gives $u_t = -\mathcal{F}(u) \in C^\theta([a + \eta, b]; H^r)$ whenever the spatial ladder has reached H^{r+4}. Its continuous representative is the time derivative, so $u \in C^1([a + \eta, b]; H^r)$. Inductively, differentiating $u_t = -\mathcal{F}(u)$ in this Banach scale and using the same estimate at each stage gives $u \in C^k([a+\eta, b]; H^r)$ after taking the spatial ladder through $H^{r+4(k+1)}$; each time derivative consumes at most four spatial derivatives. Since even r, k, and the compact interval are arbitrary, all mixed space-time derivatives exist for positive time. Thus the claimed C^∞ positive-time regularity follows without assuming a derivative above order two in the original nonlinear map.

5 Exact neutral-state result

5.1 Quadratic completion

The scalar symbol has the exact minimum

$$\mu_{\mathrm{sh}} = r - \frac{Z^2}{4Y} = \frac{26000000000947494031}{10^{20}} > 0. \tag{5.1}$$

The pinned internal matrix is

$$M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}. \tag{5.2}$$

The three leading Sylvester minors of $M - (7/250)\mathrm{I}$ are

$$\frac{9}{125}, \qquad \frac{1211}{250000}, \qquad \frac{89}{31250000}, \tag{5.3}$$

so $M > (7/250)\mathrm{I}$. Parseval therefore gives

$$\mathcal{F}_{\text{quad}}(\Psi) \geq \frac{\nu}{2} \|\Psi\|_2^2, \qquad \nu = \mu_{\text{sh}} + \frac{7}{250}. \tag{5.4}$$

The full Class-II matrix is positive definite because, after writing $P = M_X^2 + \varepsilon_M > 0$, its determinant factor is

$$c_{JJ}c_{KK} - c_{JK}^2 = \frac{1}{50} > 0. \tag{5.5}$$

For the local polynomial, direct coefficient matching gives

$$\frac{\nu}{2}\rho + \frac{\lambda}{4}\rho^2 + \frac{\gamma}{6}\rho^3 = g\rho + \frac{\gamma}{6}\rho(\rho - \rho_*)^2, \quad \rho_* = -\frac{3\lambda}{4\gamma} = \frac{43}{216}, \quad g = \frac{\nu}{2} - \frac{3\lambda^2}{32\gamma}. \tag{5.6}$$

The exact rational evaluation of g is the first value in (3.4). After integration and addition of the nonnegative Class-II term, (3.5)'s first inequality follows. Equality forces $\rho = 0$ almost everywhere, so the zero field is the unique global minimizer.

5.2 Radial derivative and absence of nonzero equilibria

The energy lower bound alone does not rule out nonzero saddles. We therefore scale $\Psi \mapsto t\Psi$ and put $y = t^2$. At a fixed field, let j denote the unscaled current and let $\ell(y) = j - q(y)\nabla\rho$ with

$$q(y) = \frac{ym}{y\rho + \varepsilon_\rho}, \qquad \theta = \frac{\varepsilon_\rho}{y\rho + \varepsilon_\rho} \in [0, 1]. \tag{5.7}$$

Direct differentiation gives

$$\frac{\mathrm{d}E_{\text{II}}}{\mathrm{d}y} = y \begin{pmatrix} j & \ell \end{pmatrix} R_\theta \begin{pmatrix} j \\ \ell \end{pmatrix}, \qquad R_\theta = \begin{pmatrix} a - \theta b & b + \theta(b-c)/2 \\ b + \theta(b-c)/2 & c(1+\theta) \end{pmatrix}. \tag{5.8}$$

For the pinned coefficients,

$$a - \theta b \geq \frac{21}{2000P} > 0, \qquad \det R_\theta = \frac{9(-81\theta^2 + 128\theta + 128)}{10240000P^2} > 0 \quad (0 \leq \theta \leq 1). \tag{5.9}$$

The numerator is concave and positive at both endpoints, so the Class-II energy is nondecreasing along every amplitude ray. Differentiating the remaining terms in the same ray gives

$$\langle D\mathcal{F}(\Psi), \Psi \rangle_{\mathbb{R}} \geq \left(\nu - \frac{\lambda^2}{4\gamma} \right) \|\Psi\|_2^2 = \kappa \|\Psi\|_2^2. \tag{5.10}$$

This is the second inequality in (3.5). A nonzero critical point would make its left-hand side vanish, contradicting (5.10). Finally,

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t} \|\Psi(t)\|_2^2 = -\langle D\mathcal{F}(\Psi(t)), \Psi(t) \rangle_{\mathbb{R}} \leq -\kappa \|\Psi(t)\|_2^2 \tag{5.11}$$

proves (3.6).

Remark 5.1 (meaning of the neutral result)**.** The result rejects nonzero equilibria only for the declared unconstrained, neutral, hash-pinned functional and its canonical gradient flow. It does not reject every retuned functional, fixed-charge problem, compact target, or conserved evolution. In particular, it is a candidate-functional rejection boundary, not a proof that a physical vacuum is empty.

6 Exact imposed-ensemble transition

6.1 The finite-torus spectral bottom

The quadratic operator has the form

$$\mathcal{L} = (r + Z(-\Delta) + Y(-\Delta)^2)\mathrm{I} + M, \quad M = \begin{pmatrix} 1/10 & -1/20 & -1/20 \\ -1/20 & 13/100 & -1/20 \\ -1/20 & -1/20 & 17/100 \end{pmatrix}. \tag{6.1}$$

The characteristic polynomial of M is

$$p(t) = t^3 - \frac{2}{5}t^2 + \frac{223}{5000}t - \frac{3}{3125}. \tag{6.2}$$

An exact Sturm computation isolates its smallest root m_* by

$$\frac{2811617233511}{10^{14}} < m_* < \frac{2811617233512}{10^{14}}. \tag{6.3}$$

For $n \in \mathbb{Z}^3$, $|k_n|^2 = \pi^2|n|^2/64$. The rational enclosure of π used by the audit proves

$$\frac{5}{2} < \frac{-Z/(2Y)}{\pi^2/64} < \frac{7}{2}. \tag{6.4}$$

Because $p(x)$ is a strictly convex quadratic in x, the last enclosure places its vertex, expressed in the integer variable $s = |n|^2$, strictly between $5/2$ and $7/2$. The unique nearest integer is therefore $s = 3$; since $3 = 1^2 + 1^2 + 1^2$, this shell exists and contains the eight body-diagonal indices $n = (\pm 1, \pm 1, \pm 1)$. Thus

$$\begin{gathered} \lambda_0 = r + Z\left(\frac{3\pi^2}{64}\right) + Y\left(\frac{3\pi^2}{64}\right)^2 + m_*, \\ 0.28811617234458494031 < \lambda_0 < 0.28811617234459494032. \end{gathered} \tag{6.5}$$

This is a finite-torus commensurability statement. It is not a BCC theorem and it does not remove the shell degeneracy in an infinite-volume limit.

6.2 Nonnegative decomposition and saturation

Set $Q = \|\Psi\|_2^2/2$. Coefficient matching for $f(\rho) = \lambda\rho^2/4 + \gamma\rho^3/6$ gives

$$\frac{\lambda_0}{2}\rho + f(\rho) = \frac{\mu_t}{2}\rho + \frac{\gamma}{6}\rho(\rho - \rho_*)^2, \quad \rho_* = -\frac{3\lambda}{4\gamma}, \quad c_* = \frac{3\lambda^2}{16\gamma}. \tag{6.6}$$

Adding the spectral and Class-II remainders yields (3.8).

Let u_* be a unit eigenvector of M for m_* and select any one of the eight indices on the shell. The plane wave

$$\Psi_*(x) = \sqrt{\rho_*}\, u_* \exp\left(\frac{2\pi i}{16} n \cdot x\right) \tag{6.7}$$

has constant ρ and constant internal bilinears. Hence $J_A = K_A = 0$, the spectral remainder vanishes, and

$$Q_* = \frac{16^3 \rho_*}{2} = \frac{11008}{27}. \tag{6.8}$$

All terms in (3.8) are therefore saturated.

6.3 Fixed charge and grand potential

For a prescribed charge Q, let $\bar{\rho} = 2Q/16^3$. The exact Bregman identity is

$$f(\rho) - f(\bar{\rho}) - f'(\bar{\rho})(\rho - \bar{\rho}) = \frac{(\rho - \bar{\rho})^2}{12}(4\gamma\bar{\rho} + 2\gamma\rho + 3\lambda). \tag{6.9}$$

If $\bar{\rho} \geq \rho_*$, the bracket is nonnegative. On the affine constraint $\int \rho = 2Q$, the linear term integrates to zero. The spectral remainder, the Class-II energy, and (6.9) consequently prove constrained global minimality of a constant-density ground-shell plane wave for all $2Q/16^3 \geq \rho_*$. The constrained problem below Q_* is not decided here. For completeness, the fixed-charge set is weakly closed in H^2: weak H^2 convergence has a strong L^2 subsequence on the fixed torus, hence $Q(\Psi_n) \to Q(\Psi)$.

Subtracting μQ from (3.8) gives

$$\mathcal{O}_\mu(\Psi) = (\mu_t - \mu)Q(\Psi) + \frac{1}{2}\langle \Psi, (L - \lambda_0)\Psi\rangle_{\mathbb{R}} + \mathcal{F}_{\mathrm{II}}(\Psi) + \frac{\gamma}{6}\int \rho(\rho - \rho_*)^2\,\mathrm{d}x. \tag{6.10}$$

For $\mu < \mu_t$, every nonzero field has positive grand potential. At $\mu = \mu_t$, zero and the saturated states coexist. For $\mu > \mu_t$, the plane wave has negative grand potential. We now make the existence step in this last assertion explicit. Write $\delta = \mu - \mu_t > 0$. The lower bound (4.3) and positivity of the internal matrix give the following explicit high-frequency estimate:

$$\frac{1}{2}\langle \Psi, (L - \lambda_0)\Psi\rangle_{\mathbb{R}} \geq \frac{1}{10}\|\Psi\|^2_{H^2_\Delta} - \frac{\lambda_0}{2}\|\Psi\|_2^2. \tag{6.11}$$

The scalar polynomial also obeys, for $\rho \geq 0$,

$$\frac{\gamma}{6}\rho(\rho - \rho_*)^2 - \frac{\delta}{2}\rho \geq \frac{\gamma}{12}\rho^3 - C_\delta, \tag{6.12}$$

by Young's inequality. On the finite torus, the L^2 term in (6.11) is absorbed by the L^6 term in (6.12) up to a constant. Thus $\mathcal{O}_\mu$ is bounded below and coercive in H^2. If a minimizing sequence Ψ_n converges weakly in H^2, Rellich compactness gives strong convergence in H^1. More precisely, on the three-torus $H^2 \Subset C^{0,\alpha}$ for every $\alpha < 1/2$, so (after a subsequence) $\Psi_n \to \Psi$ uniformly. Hence the local potential is continuous, while the spectral quadratic form is weakly lower semicontinuous. Writing u_n and u for the realifications of Ψ_n and Ψ, the coefficients $B(u_n)$ in (4.8) converge uniformly to $B(u)$ and $\nabla u_n \rightharpoonup \nabla u$ in L^2. The positive quadratic Class-II integral is therefore weakly lower semicontinuous as well (equivalently, use the uniformly convergent positive square roots $B(u_n)^{1/2}$). The direct method therefore supplies a global minimizer; its value is negative because of the plane wave, so it is necessarily nonzero.

The zero-field Hessian loses positivity at $\mu_{\mathrm{lin}} = \lambda_0$. Along the ground-plane-wave amplitude family, stationary densities solve

$$\gamma\rho^2 + \lambda\rho + (\lambda_0 - \mu) = 0. \tag{6.13}$$

The amplitude saddle-node is $\mu_{\mathrm{sn}} = \lambda_0 - \lambda^2/(4\gamma)$, while coexistence is $\mu_t = \lambda_0 - 3\lambda^2/(16\gamma)$. With the pinned coefficients this is exactly (3.10). At $\mu = \mu_t$, the saturated plane-wave minimizer has charge Q_*, while zero is also a minimizer and remains a strict local minimum because $\mu_t < \mu_{\mathrm{lin}}$. For $\mu > \mu_t$, the larger-amplitude plane-wave stationary root has $\rho > \rho_*$, but the direct-method argument above does not prove that this branch is the unique global minimizer or identify the charge of every global minimizer. Thus the theorem establishes finite-volume grand-potential coexistence between two explicitly exhibited charges, 0 and Q_*, at the coexistence point; a full one-sided global-charge discontinuity theorem would require an additional selection or stability argument.

Remark 6.1 (why R-157 and R-158 do not conflict)**.** At the nonzero state (6.7), the constrained Euler–Lagrange equation is $D\mathcal{F}(\Psi_*) = \mu_t\Psi_*$, not $D\mathcal{F}(\Psi_*) = 0$. R-157 concerns the latter equation in the unconstrained linear space; R-158 changes the variational problem. Moreover, $\mathcal{F}(\Psi_*) = \mu_t Q_* > 0 = \mathcal{F}(0)$, so the ordered state does not beat the original neutral reference.

7 Relation to prior work and residual contribution

The functional-analytic ingredients used here are standard: sectorial fourth-order semigroups, Sobolev product estimates, Galerkin compactness, weakly singular Gronwall inequalities, and polynomial coercivity. A Swift–Hohenberg analysis by Giorgini [3] proves well-posedness and long-time attractor properties for scalar slow/fast equations, but does not contain the present three-component regularized Class-II map, its exact radial sign, or the imposed-ensemble completion. Thus the well-posedness theorem alone is not claimed as a new general method.

The abstract framework of Asai [4] treats fourth-order quasilinear parabolic equations by continuous maximal regularity. It is relevant background, but its hypotheses do not directly identify the present problem: here the principal fourth-order operator is constant, while the regularized Class-II contribution is a model-specific order-two lower-order map. We therefore verify the required Sobolev and chain-rule estimates explicitly rather than importing an existence theorem without a crosswalk.

The closest current quasilinear pattern-formation comparisons are two results of Belin and Schneider. Their reaction–diffusion–advection theorem [5] justifies a Ginzburg–Landau approximation near a first instability, and their later uniformly-local theory [6] applies a related argument to the one-dimensional fully nonlinear Swift–Hohenberg model with nonlinearity $-\partial_x^4(u^3)$. In the latter theorem, solutions remain close to a small-amplitude modulation on $0 \le t \le T_0/\varepsilon^2$ in uniformly local Sobolev spaces. These works show that quasilinear Swift–Hohenberg analysis or maximal-regularity handling is not by itself a residual novelty here. They do not, however, give an all-data global H^2 gradient flow on a three-dimensional torus for the present three-component functional, nor the neutral radial rejection and imposed-ensemble identities proved below.

Two coupled-system comparisons further delimit this statement. Hilder and Kuehn [7] rigorously derive coupled complex Ginzburg–Landau equations near a resonant Turing/Turing–Hopf instability, prove finite-long-time approximation estimates, and construct selected global spatially periodic solutions for a coupled Swift–Hohenberg system. This is a substantially closer coupled-PDE comparison, but it is a near-instability reduction and solution-construction theorem rather than all-data global well-posedness for the present gradient flow. Becker et al. [8] study pattern coexistence and interface control in two anti-symmetrically coupled one-dimensional Swift–Hohenberg equations by weakly nonlinear analysis and simulation. Their coexistence is spatial competition between dynamically stable patterns, not the exact equality of global grand-potential minimizers asserted in Theorem 3.3.

The original Brazovskii analysis [9] and the continuous-shell study of Swift and Leitner [10] provide the physical context for nonzero-wavevector minima and fluctuation-driven first-order transitions. They do not prove the exact finite-torus algebraic completion used here. In a closer mathematical PFC setting, Martine-La Boissoniere, Choksi, and Lessard [11] use rigorously validated numerics for scalar two-dimensional steady states and local minimizers; their conserved H^{-1} flow, dimension, and target statements differ from the analytic three-component L^2 flow and finite-shell ensemble theorem proved here.

Recent computer-assisted work by Duchesne, Lessard, and Takayasu [12] proves global existence and convergence for scalar two- and three-dimensional Swift–Hohenberg problems, and treats derivative nonlinearities in an Ohta–Kawasaki example. Its validated semilinear setting and equilibrium-specific trapping arguments are adjacent but do not subsume the present regularized three-component Class-II map or its exact neutral and ensemble identities. Ruan [13] gives a self-contained constrained Landau–Brazovskii periodic minimizer; the field, variational reduction, and conclusion differ from the six-real- component fourth-order flow and the grand-potential coexistence theorem here. For multicomponent PFC models, Bao, Chen, and Jiang [14] analyze a discretized coupled-mode Swift–Hohenberg optimization algorithm with binary, ternary, and quinary numerical tests; this is algorithmic finite-dimensional convergence rather than the present continuum analytic theorem. Finally, Mi, Cui, and You [15] study one-dimensional complex Swift–Hohenberg periodic and quasi-periodic solutions by KAM methods, not the declared real gradient-flow or imposed-charge problem. These comparisons sharpen the non-subsumption boundary but are not a priority claim.

Quantum anharmonic-crystal phase-transition work, such as Kargol–Kondratiev–Kozitsky [16] develops Euclidean Gibbs states, reflection arguments, and infrared estimates for specified lattice models. The present paper does not import a quantum phase theorem. It instead treats a classical fixed-torus variational functional and proves the model-specific sign and spectral identities needed for its two ensembles. The residual contribution is therefore not a new general quasilinear method, but the exact combination of (a) the all-data derivative Class-II well-posedness argument for this pinned functional, (b) the global neutral rejection, and (c) the finite-shell ensemble classification, all with a common functional and an explicit scope firewall.

The claim is not a world-first assertion. A complete novelty determination requires a specialist literature search beyond the bounded crosswalk recorded in `literature-crosswalk.md` and independent mathematical review of the model-specific estimates.

8 Verification and reproducibility

The paper cites only the registered A2 claim and its result records:

component	independent executable evidence	recorded result
full evolution	four audits	20/20, 14/14, 12/12, 15/15 PASS
R-157 neutral sign	primary and non-importing lanes	26/26, 24/24 PASS
R-158 ensemble sign	primary and standard-library lanes	35/35, 24/24 PASS
R-472 exact core	Lean sidecar and hostile checks	30/30, 22/22, 12/12, 22/22 PASS

The canonical commands are listed in `verification/README.md`. The registered A2/H3 record is used for provenance and transfer only; the theorem checks below are run against the explicit functional printed in this paper. The tests recompute rational constants, matrix signs, spectral intervals, normalizations, and hostile mutations. They do not prove the analytic semigroup, compactness, chain-rule, or external-literature steps; those remain the proof text's responsibility.

The two blank, package-local proof and novelty review contracts map every requested disposition to Theorems 3.1–3.3, exact equation labels, frozen source hashes, and dependent-item rules. Their presence records no review outcome.

The 2026-09-03 replay passed the A2 wrapper and all four R-157/R-158 child lanes. After the structured T-054 predicate repair recorded in EXP-001372, the R-157 integrated lane passes 144/144 (legacy A2 61/61) and the R-158 integrated lane passes 155/155 with the R-157/A2 regression. This closes the repository synchronization gate only; the repository release check is already passing, while the paper remains at draft pending the analytic proof audit, specialist literature review, operator confirmation, and capstone bundle.

The existing A2 T6 reproduction bundle is recorded in the claim card and its bundle directory. A dedicated capstone bundle containing the integrated R-157/R-158 referee note and all transitive scripts is a required post-review action. Under the repository policy it must be built only after operator confirmation of the integrated referee package. This draft does not claim that confirmation or a release-gated submission.

9 Limitations and falsifiers

The theorem's domain is the fixed side-16 torus, six-real-component H^2 field space, pinned coefficients, positive density floor, and $\eta_{\mathrm{shell}} = 0$. The following are outside its scope:

- infinite-volume or continuum removal, counterterm flow, and regulator independence;
- a real-time infinite-volume dynamics, algebraic KMS states, ground states, spectral gaps, extremality, purity, or clustering;
- a conserved physical charge, a reservoir, or a derivation of μ;
- gauge completion, gauge-invariant density modulation, BCC or unique morphology;
- the historical non-variational backend, signed stochastic composites, alternative parameters, and compact target spaces;
- any physical-vacuum, gravity, cosmology, or Sector-A closure claim.

The analytic theorem is falsified by a counterexample to global H^2 well-posedness, uniqueness, continuous dependence, the exact energy identity, or positive-time smoothing in the declared domain. R-157 is falsified by an exact nonzero critical point or a field violating (3.5). R-158 is falsified by failure of the exact spectral shell, the decomposition, the Bregman sign, plane-wave saturation, or the ordering (3.10). A failure of the named P1 identification hypothesis blocks transfer to the TECT production interpretation but does not alter the mathematics of the explicitly defined functional.

10 Conclusion

The pinned functional has a sharp two-sided mathematical verdict. Its neutral unconstrained dynamics are globally well posed and converge exponentially to zero; no nonzero equilibrium survives the exact radial sign. A nonzero shell state appears only after the variational problem is changed by an imposed charge or chemical potential, where an exact first-order coexistence mechanism can be proved on the finite torus. This separation is the principal result: the ensemble mechanism is mathematically real, while its physical provenance and interpretation remain open. The manuscript is therefore a candidate independent mathematical paper, currently at draft status pending the specialist novelty review, analytic proof audit, operator confirmation, and capstone bundle specified in the accompanying files; the repository release check is a separate passing consistency gate.

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